

# Benjamin Schaffer

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## Professional Summary

Mechanical Engineering student with hands-on experience in mechanical design, manufacturing processes, and systems integration. Proven leader on large, cross-functional engineering teams, with a strong foundation in CAD, prototyping, and embedded systems. Seeking an engineering internship to apply technical and leadership skills to real-world engineering challenges.

## Leadership Experience

**Project Manager – MIE 375 Manufacturing Processes** *University of Massachusetts Amherst | Sep 2025 – Dec 2025*

- Led a **40-member** enterprise across **8 BUs**; ran stand-ups/design reviews and ensured **QMS** conformance
- Delivered a **PCB-mount assembly** upgrade for medical imaging; managed requirement changes and drove **DFMA** updates

**Technical Lead – Collegiate Wind Competition Team** *University of Massachusetts Amherst | Sep 2025 – Present*

- **Manufacturing Design & Materials Lead** for the prototype turbine; select materials/processes to optimize weight/strength
- Lead a sub-team of **8 members**, managing task assignments and technical deliverables
- Ensure compliance with size/transport limits & rules; design jigs/fixtures and a **repeatable build/test plan**

**Chapter Representative – Engineers Without Borders** *University of Massachusetts Amherst | Feb 2025 – Present*

- **Fundraising Lead** for the EWB local project; plan auctions and 6k runs, manage outreach and donor communications
- **Secured \$500** in project funding by placing 1<sup>st</sup> out of 14 applicants in the **CLACLS Undergraduate Community Project Award**

## Engineering Projects

**Wrench Design – MIE 313 Design of Mechanical Components** *University of Massachusetts Amherst | Sep 2025 – Dec 2025*

- Designed an ultralightweight bicycle wrench using **SOLIDWORKS & ANSYS**; reducing mass while increasing safety factor
- Fabricated the final design using manual machine shop methods, including milling, cutting, and finishing operations

**Competitive Robot Design – ASME Student Design Team** *University of Massachusetts Amherst | Sep 2023 – May 2024*

- **Designed and built** a competitive robot using **SOLIDWORKS & CAD modeling**
- Integrated **mechanical and electrical systems** with a team of engineers for autonomous functionality

**Greenhouse Volunteer Projects – Engineers Without Borders** *University of Massachusetts Amherst | Feb 2025 – Present*

- Develop a low-cost biogas heating system; perform structural assessment/reinforcement of a deteriorated greenhouse.

**Autonomous Vehicle Design – ECE 361 Electrical Engineering** *University of Massachusetts Amherst | Sep 2024 – Dec 2024*

- Implemented **obstacle detection** and avoidance algorithms using ultrasonic sensors on an **Arduino-powered** vehicle
- Applied **embedded systems** programming (C/C++) and real-time **sensor integration** to build a self-navigating robot

**Quadcopter UAV – Independent Project** *Jul 2025 – Aug 2025*

- Built a quadcopter UAV incorporating a **3D-printed** frame, **onboard electronics**, and stable **flight control**
- Integrated camera, LED lighting, and explored FPV; developed skills in **aerodynamics** and **wireless control**

## Professional Experience

**Systems & Technology Intern** *Westford Public Schools, Westford, MA | Jun 2024 – Aug 2024 & May 2025 – Sep 2025*

- Prepared and updated **thousands of school devices**, performing software resets and system updates
- Diagnosed and repaired **hardware issues**, providing on-site technical support across multiple locations

**Customer Service & Sales Associate** *UPS, Westford, MA | Aug 2021 – May 2022*

- Processed **customer transactions** with accuracy and efficiency in a high-volume retail setting
- Managed **returns, exchanges, and refunds**, ensuring compliance with store policies and customer satisfaction

## Technical Skills

- **CAD, FEA, and mechanical design:** SOLIDWORKS, ANSYS, technical drawings, structural simulation
- **Programming and Analysis:** MATLAB, Simulink, and Python for engineering Analysis
- **Prototyping and Embedded Systems:** Arduino-based automation, robotic design, rapid prototyping
- **Digital Fabrication and Tools:** 3D Printing & Slicing (Cura), Microsoft Office, Google Workspace

## Education

**University of Massachusetts Amherst, Amherst, MA** *Bachelor of Science in Mechanical Engineering, Minor in Business*

- **GPA: 3.85 | Commonwealth Honors College | Dean's List Honoree**
- **Relevant Coursework:** Physics, Calculus, Thermodynamics, Materials Science, Statics & Dynamics, Mechanics of Materials, Fluid Mechanics, Design of Mechanical Components, Design of Mechanical Assemblies, Manufacturing Processes, Dynamic Systems, Heat Transfer, Vehicle Automation, Technical Writing, Engineering Professionalism
- **Extracurriculars:** ASME Student Design Team & General Body, Engineers Without Borders, Collegiate Wind Competition Team

## Languages

- **English** (Native)
- **German** (Intermediate)